

# Irrationality of Cubic Threefolds after One Stabilization

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## Abstract

We prove that  $X \times \mathbf{P}^1$  is irrational for every smooth complex cubic threefold  $X$ . Our invariant counts certain rank-two blocks of the generic even quantum  $D$ -module. A counted block has centered leading term  $N \neq 0$  with  $N^2 = 0$ , and after a canonical elementary modification its two residue eigenvalues are distinct modulo  $\mathbf{Z}$ . Iritani's blowup decomposition and weak factorization make this count birationally invariant because every point, curve, and surface center contributes zero. Iritani–Koto's projective-bundle decomposition computes its value after stabilization; the resulting count distinguishes  $X \times \mathbf{P}^1$  from  $\mathbf{P}^4$ . The same count gives an irrationality criterion for smooth projective threefolds, the one-stabilization theorem for prime Fano threefolds of genus eight, and irrational universally  $CH_0$ -trivial fourfolds.

**Keywords.** cubic threefold; universal  $CH_0$ -triviality; irrationality; quantum  $D$ -modules; formal exponents; birational invariants; weak factorization; Fano threefolds.

## 1 Introduction

Let  $X \subset \mathbf{P}^4$  be a smooth complex cubic threefold. Clemens and Griffiths proved that  $X$  is irrational [6], but their intermediate-Jacobian argument does not directly obstruct the rationality of  $X \times \mathbf{P}^1$ : in a weak factorization of a birational map between smooth fourfolds, a surface center  $S$  contributes  $H^3(S)(-1)$  in degree five, while  $H^3(X)(-1)$  occurs in  $H^5(X \times \mathbf{P}^1)$ . We prove that the stabilized fourfold is nevertheless irrational.

**Theorem 1.1** (One-step irrationality). *For every smooth complex cubic threefold  $X$ , the fourfold  $X \times \mathbf{P}^1$  is irrational.*

The proof uses the generic even quantum  $D$ -module (QDM). After extending its generic coefficient field to an algebraic closure, the generalized eigenspaces of quantum multiplication by the Euler field decompose the formal connection into blocks. We record their isomorphism classes in a free commutative monoid. For an additive marker compatible with the QDM comparisons, Iritani's blowup theorem gives a formula for its change under blowup, with the individual center summands retained separately. Combined with weak factorization, this formula makes any such marker that vanishes on every possible center in dimension four a birational invariant. Iritani–Koto's projective-bundle theorem computes the marker under product with projective space.

The marker counts rank-two blocks for which the centered leading term  $N$  of Euler multiplication is nonzero and satisfies  $N^2 = 0$ . Horizontality of the Poincaré pairing makes a canonical elementary modification regular singular, while flatness gives a Lax equation for its residue and hence preserves its eigenvalues. We count the block when the two residue eigenvalues, equivalently the two formal exponent classes, are distinct modulo  $\mathbf{Z}$ . Using Beauville's quantum product formulas, we compute exponent representatives  $-1/6$  and

$-5/6$  for the cubic block, with difference  $2/3$ . The resulting marker vanishes on points, curves, and surfaces, doubles on product with  $\mathbf{P}^1$ , and vanishes on projective space. This proves the theorem.

The comparison inputs are Iritani’s blowup decomposition and Iritani–Koto’s projective-bundle decomposition [8, 9]. The new point is to combine these formulas with weak factorization while retaining the individual center summands, and then to apply an additive rank-two exponent marker that vanishes on every center that can occur in dimension four. Cai independently computes the cubic quantum differential equation in a normalization compatible with ours [5]; the birational marker and stabilization argument are not part of that calculation.

The same construction gives a threefold criterion without stabilization.

**Corollary 1.2** (Threefold marker criterion). *Let  $Y$  be a smooth projective complex threefold. If its generic even QDM contains a block marked by (2.12), then  $Y$  is irrational.*

Indeed, the only centers in a weak factorization of threefolds have dimension at most one, where the marker vanishes, while the generic quantum product of  $\mathbf{P}^3$  is semisimple. This gives an irrationality criterion formulated in terms of the quantum  $D$ -module. We do not claim that this marker coincides with a semiorthogonal or derived-categorical rationality obstruction.

A further application comes from Kuznetsov’s construction. For an arbitrary smooth prime Fano threefold  $V$  of genus eight, it associates a smooth Pfaffian cubic threefold  $X$  and rank-two projective bundles over  $V$  and  $X$  related by a flop [10, Theorems 2.17–2.18]. This gives the following consequence.

**Corollary 1.3** (Irrationality after one stabilization for  $V_{14}$ ). *For every smooth prime Fano threefold  $V$  of genus eight,  $V \times \mathbf{P}^1$  is irrational.*

The theorem also produces universally  $CH_0$ -trivial fourfolds that are irrational. Universal  $CH_0$ -triviality passes to a projective bundle by the projective-bundle formula for Chow groups.

**Corollary 1.4** (Separation in codimension at most three). *There is a nonempty countable union of subvarieties of codimension at most three in the moduli space of smooth cubic threefolds such that, for every member  $X$ , both  $X$  and  $X \times \mathbf{P}^1$  are universally  $CH_0$ -trivial, while  $X \times \mathbf{P}^1$  is irrational.*

In the next statement, the displayed hypersurface lies in  $\mathbf{P}^4$  and is the Fermat cubic threefold.

**Corollary 1.5** (Fermat cubic). *The fourfold*

$$\{x_1^3 + x_2^3 + x_3^3 + x_4^3 + x_5^3 = 0\} \times \mathbf{P}^1$$

*is universally  $CH_0$ -trivial and irrational.*

**Corollary 1.6** (Coprime-degree cubics). *Let  $X$  be one of the cubic threefolds of [13, Theorem 3.3]. Then  $X \times \mathbf{P}^1$  admits unirational parametrizations of coprime degrees, is universally  $CH_0$ -trivial, and is irrational.*

Voisin proves the codimension-at-most-three statement [12, Theorem 4.5]. The Fermat case follows from Colliot-Thélène’s separated-variable criterion [7]. Yang–Yu–Zhu construct the coprime-degree family [13, Theorems 1.1 and 3.3]. The last corollary answers the  $m = 1$  case of their question; it does not address all stabilizations.

Section 2 constructs the block monoid and its QDM operation formulas, defines the exponent marker, computes the cubic block, and proves that the marker vanishes on centers of dimension at most two. Section 3 gives the threefold, genus-eight, and universal- $CH_0$  consequences.

## 2 QDM markers and one stabilization

We prove the theorem by applying the quantum  $D$ -module decomposition formulas directly. The common argument is stated first for an arbitrary additive marker. The rest of the section constructs the smallest marker needed for a cubic threefold and proves that it vanishes on every possible low-dimensional center.

Here is the concrete invariant before the general construction. Split the generic even quantum  $D$ -module into formal connection blocks. A block contributes one when it has rank two, its centered leading Euler operator is nonzero square-zero, and the two exponent classes obtained after the canonical elementary modification are distinct modulo  $\mathbf{Z}$ ; every other block contributes zero. Their sum is  $I_{\text{at}}$ . The rest of the section shows that this count is intrinsic and additive, vanishes on all possible centers, equals one on a cubic threefold, doubles after product with  $\mathbf{P}^1$ , and vanishes on  $\mathbf{P}^4$ .

### 2.1 The one-blowup model

Let  $Y$  be smooth and projective. Pass to numerical Novikov variables, set the odd bulk variables to zero, and restrict the carrier of the quantum connection to  $H^{\text{ev}}(Y)$ . The last restriction is essential: even bulk alone still leaves the odd cohomology in the fibre. Let  $K_Y$  be the generic coefficient field of this even QDM and fix an algebraic closure  $\bar{K}_Y$ . All spectral blocks below are formed only after base change to  $\bar{K}_Y$ . Thus later scalar extensions do not split a primary factor that was previously counted as one block.

Let  $U$  be multiplication by the Euler field on the even carrier. Its generalized eigenspaces over  $\bar{K}_Y$  are summands of the full formal connection, rather than merely summands of  $U$ :

**Proposition 2.1** (Formal spectral connection splitting). *Suppose the leading operator of a finite free formal QDM has block-diagonal form  $U_i = \lambda_i I + N_i$ , where each  $N_i$  is nilpotent and  $\lambda_i - \lambda_j$  is a unit for  $i \neq j$ . There is a unique gauge  $G = I + O(z)$ , regular in  $z$  and block off-diagonal in positive orders, which makes the  $z$ -connection block diagonal. In this gauge every bulk and Novikov direction is block diagonal as well. The horizontal pairing is block diagonal and nondegenerate on each factor.*

*Proof.* At order  $z^{n-1}$ , the  $(i, j)$ -entry of the gauge equation is a Sylvester equation

$$U_i X - X U_j = B_{n,ij}.$$

Its operator is  $(\lambda_i - \lambda_j)I$  plus the nilpotent operator  $X \mapsto N_i X - X N_j$ , hence is invertible. Recursion gives the normalized gauge, and the same equation with zero right-hand side proves uniqueness.

Choose frames of the generalized-eigenspace subbundles after the splitting extension. If an off-diagonal block of a base-direction connection had a first nonzero Laurent coefficient  $Xz^m$ , the lowest off-diagonal coefficient of flatness would be  $U_i X - X U_j = 0$ , contrary to Sylvester invertibility. Thus every base direction preserves the same factors. Applied coefficientwise to pairing horizontality, the identical argument kills the off-diagonal pairing blocks. Nondegeneracy of the total pairing then makes every diagonal restriction nondegenerate.  $\square$

We call these factors the *generic even QDM blocks* of  $Y$ . A basic block retains its connection, pairing, and centered Euler endomorphism, but not an absolute cohomological grading. A block type may require additional marked data. If it retains grading, morphisms may carry a recorded integral suspension and its marker fold must be invariant under that suspension. The atomic type below forgets absolute grading. Its morphisms preserve all retained data and are regular connection isomorphisms after scalar extension and formal change of bulk coordinates. In particular, their gauges use neither negative nor fractional powers of the connection variable  $z$ .

Fix a block type  $\mathcal{T}$ . Let  $\Pi_{\mathcal{T}}$  be the regular-isomorphism groupoid of blocks of that type and put

$$\mathcal{L}_{\mathcal{T}} := \pi_0 \operatorname{Sym}^{\sqcup}(\Pi_{\mathcal{T}}),$$

the effective free commutative monoid on their isomorphism classes. Direct sum gives each  $Y$  a block ledger  $\mathcal{B}_{\mathcal{T}}(Y) \in \mathcal{L}_{\mathcal{T}}$ . A *marker fold* is a commutative-monoid homomorphism

$$w : \mathcal{L}_{\mathcal{T}} \longrightarrow A, \quad I_w(Y) := w(\mathcal{B}_{\mathcal{T}}(Y)),$$

where  $A$  is an effective commutative monoid. No group completion is made: it can destroy useful folds, including Boolean ones.

The two specializations use the following notation.

| Symbol                                        | Meaning                                                                             |
|-----------------------------------------------|-------------------------------------------------------------------------------------|
| $\Pi_{\mathcal{T}}$                           | regular-isomorphism groupoid of typed QDM blocks                                    |
| $\mathcal{L}_{\mathcal{T}}$                   | $\pi_0 \operatorname{Sym}^{\sqcup}(\Pi_{\mathcal{T}})$ , the effective block monoid |
| $\mathcal{B}_{\mathcal{T}}(Y)$                | block ledger of $Y$                                                                 |
| $w : \mathcal{L}_{\mathcal{T}} \rightarrow A$ | additive marker fold into an effective monoid                                       |
| $\omega_j$                                    | the $j$ -th center occurrence in one blowup                                         |
| $c_w(\omega_j)$                               | marker of that actual center occurrence                                             |

Table 1: Notation for the occurrence-indexed marker ledger.

Consider one blowup  $\beta : \tilde{Y} = \operatorname{Bl}_Z Y \rightarrow Y$  along a smooth center of codimension  $c \geq 2$ . Iritani's comparison has one ambient summand and  $c - 1$  center summands. We retain the occurrence data of each center summand: for the  $j$ -th, record

$$\omega_j = (\beta, Z, j, \varsigma_j, \chi_j), \quad 0 \leq j \leq c - 2,$$

where  $\varsigma_j$  is Iritani's center reconstruction parameter and  $\chi_j$  is the induced numerical Novikov specialization. Its correction is  $c_w(\omega_j) := w(\mathcal{B}_{\mathcal{T}}(Z; \omega_j))$ . Equality is required only after applying  $w$ ; distinct block multisets may have the same marker.

**Theorem 2.2** (Occurrence-indexed marker ledger). *Let  $w : \mathcal{L}_{\mathcal{T}} \rightarrow A$  be a marker fold unchanged by the allowed regular scalar extensions and formal coordinate changes. Suppose*

$$I_w(\mathbf{P}_Y(V)) = rI_w(Y), \quad \operatorname{rk} V = r, \quad (2.1)$$

$$I_w(\operatorname{Bl}_Z Y) = I_w(Y) + \sum_{j=0}^{c-2} c_w(\omega_j), \quad \operatorname{codim}_Y Z = c \geq 2. \quad (2.2)$$

*If every actual smooth center occurrence of dimension at most  $d - 2$  has zero correction, then  $I_w$  is a birational invariant of smooth projective  $d$ -folds.*

*Proof.* Projective weak factorization writes a birational map of smooth projective  $d$ -folds as a zigzag of blowups and blowdowns along smooth centers of codimension at least two [1, Theorem 0.3.1]. By (2.2) and center vanishing, the marker is unchanged at each step; the equalities therefore telescope. Disconnected centers are treated componentwise, while a divisor blowup is an isomorphism.  $\square$

Equivalently, the marker descends on object sets:

$$\begin{array}{ccc} \mathrm{Ob}(\mathrm{SmProj}_d) & \xrightarrow{I_w} & A \\ \downarrow & & \parallel \\ \mathrm{Ob}(\mathrm{SmProj}_d)/\sim_{\mathrm{bir}} & \xrightarrow{\bar{I}_w} & A. \end{array} \quad (2.3)$$

This square reformulates the proof; it is not an additional input.

## 2.2 The QDM operation providers

Four compatibility facts place the comparison theorems in the preceding framework. First, the intrinsic ample-adic completion and the Laurent neighborhood of a fibre or exceptional cusp need not map to one another. Iritani's display (1.1) supplies the required common coefficient maps [8, (1.1)]; injectivity for each center occurrence is established below. For projective bundles the common source is  $\mathbf{C}[z, q][[Q, \hat{\tau}]]$ . On the blowup side the ambient term uses the corresponding graded source. A center term begins with its intrinsic reduced coefficient ring from Remark 2.3, maps to the possibly smaller image ring of Remark 5.6, and is then pulled to a faithfully tagged occurrence ring in the external direct sum. Embed the generic fields of the ambient common image and of these faithfully tagged occurrence images in a fixed algebraically closed overfield and form every primary block there. Base change after this splitting extension preserves the rank, nilpotent Jordan type, regular formal gauge, and formal exponent classes of each block; forming primary factors over a nonsplitting field before extension would not.

Second, the comparison is restricted to the even carrier, not merely to the even bulk base. The supermodule conventions and the pull-push and Fourier formulas in Iritani's Section 5.8 and Iritani–Koto's construction are parity-preserving. After the odd bulk variables are set to zero, the comparison and its inverse therefore restrict to mutually inverse maps on  $H^{\mathrm{ev}}$ . This is the QDM on which the atomic ledger is formed.

Third, the comparisons are regular in  $z$ . Iritani's Theorem 5.18 and Iritani–Koto's Theorem 5.1 are respectively defined over rings of the form

$$\mathbf{C}[z]((q_{\mathrm{exc}}^{-1/s}))[[Q, \hat{\tau}]], \quad \mathbf{C}[z]((q_{\mathrm{fb}}^{-1/r'}))[[Q, \hat{\tau}]],$$

where Iritani–Koto take  $r' = r$  when  $r - 1$  is even and  $r' = 2r$  when  $r - 1$  is odd. Here  $\mathfrak{s} \in \{c - 1, 2(c - 1)\}$  is Iritani's parity-dependent ramification index from [8, (5.11)]. Their inverses use no negative powers of  $z$ ; see also [9, Remark 5.3]. The maps are homogeneous but need not have degree zero: the projective-bundle map has degree  $-(r - 1)$ , while Iritani's ambient and center components have degrees 0 and  $-c$ , respectively [9, Theorem 5.1(3)][8, Theorem 5.18(3)]. Because the atomic block type does not retain absolute grading, these shifts do not change its component. For a graded refinement they are recorded as integral suspensions, and the operation formula applies only to a fold invariant under those suspensions.

Fourth, Iritani's center map need not be injective [8, (5.15)]. After numerical reduction it has the form

$$Q_Z^d \longmapsto Q^{i_* d} q_{\mathrm{exc}}^{-\rho_Z \cdot d / (c - 1)}. \quad (2.4)$$

Here  $\rho_Z = c_1(N_{Z/Y})$ . The raw map is not the coefficient map used below. Remark 2.3 gives the intrinsic divisor-equation-reduced coefficient ring, whose Novikov generators are the combined symbols

$$X_d = Q_Z^d \exp(\sigma^{(2)} \cdot d);$$

the divisor coordinates  $\sigma^{(2)}$  are not separate elements of this ring. Remark 5.6 places its pushforward under the raw Novikov map over the image of that reduced ring [8, Remarks 2.3 and 5.6]. In the external direct sum of Section 5.8.2, by contrast, the variables  $s_j \in H^*(Z)$  are independent coordinates on the common comparison target, and the center parameter is identified as  $\varsigma_j = \varsigma_j^\circ + s_j$  [8, (5.47)–(5.48)]. The divisor components of these *target-only* coordinates retain the curve-class information lost by (2.4).

**Lemma 2.3** (Faithful center base change). *Let  $R_{Z,\text{src}}^{\text{red}}$  be the coefficient domain whose  $\mathbf{C}[z]$ -extension is Iritani's divisor-equation-reduced source ring  $\mathbf{C}[z][[Q_Z e^{\sigma^{(2)}}, \sigma']][\sigma^0]$ . Let  $A_j$  be the numerical coefficient domain of the external direct sum in Section 5.8.2, with the independent target coordinates  $t, s_0, \dots, s_{c-2}$  and with  $z$  omitted. The pullback determined by the center Novikov map and  $\varsigma_j = \varsigma_j^\circ + s_j$  gives a continuous homomorphism*

$$\iota_j : R_{Z,\text{src}}^{\text{red}} \longrightarrow A_j.$$

On a combined divisor-equation generator it is

$$X_d = Q_Z^d \exp(\sigma^{(2)} \cdot d) \longmapsto Q^{i_* d} q_{\text{exc}}^{-\rho_Z \cdot d / (c-1)} \exp((\varsigma_j^{\circ, (2)} + s_j^{(2)}) \cdot d). \quad (2.5)$$

This map is injective. In particular, it induces a field embedding

$$\text{Frac}(R_{Z,\text{src}}^{\text{red}}) \hookrightarrow \text{Frac}(A_j).$$

After this field extension, each center summand in Iritani's comparison is the faithful scalar extension, followed by the formal coordinate change  $\varsigma_j$ , of the intrinsic generic numerical even QDM of  $Z$ .

*Proof.* Choose integral divisors  $D_1, \dots, D_\rho$  whose pairings separate  $N_1(Z)$ , and use their dual coordinates in  $s_j^{(2)}$ . The variables  $s_j$  occur only in the target  $A_j$ , not as coefficients in  $R_{Z,\text{src}}^{\text{red}}$ . This asymmetry is essential.

Filter the source by an ample degree pulled back from  $Y$ , together with the ordinary bulk order. Every fibre of the untagged monomial map in a bounded piece is finite: the restriction of an ample divisor from  $Y$  is ample on  $Z$ , and its bounded slice of  $\overline{NE}(Z)$  contains only finitely many lattice points. This filtration is respected by the target map because  $H \cdot i_* d = (i^* H) \cdot d$ . The fixed shift  $\varsigma_j^\circ$  and the translations of the nondivisor and unit coordinates are invertible formal changes of target coordinates, so they do not affect injectivity.

Suppose a nonzero source element maps to zero and take its first nonzero filtered part. Group its finitely many terms by their common untagged target monomial and by their nondivisor bulk monomial. On one nonzero group the remaining relation has the form

$$\sum_{d \in F} a_d \exp\left(\sum_{k=1}^{\rho} (D_k \cdot d) s_{j,k}\right) = 0, \quad a_d \neq 0,$$

over a characteristic-zero coefficient domain that does not contain the variables  $s_{j,k}$ . The exponent vectors are distinct. Choose an integral one-parameter restriction on which they remain distinct. Derivatives of orders  $0, \dots, |F| - 1$  at the origin give a Vandermonde matrix,

hence every  $a_d$  is zero, a contradiction. Thus  $\iota_j$  is injective on every bounded quotient and, by separated completeness, on the full reduced source.

Remark 2.3 of [8] gives the intrinsic reduced source, and Remark 5.6 gives its image under the noninjective map (5.15). Equations (5.47)–(5.48) then pull that image into the external direct sum over  $A_j$ , where the variables  $s_j$  are independent, and identify the direct sum with the blowup module. Associativity of scalar extension therefore identifies the center summand with scalar extension directly along the injective composite  $\iota_j$ . Passing to fraction fields is faithfully flat. Finally, the inverse of the combined formal coordinate change  $(\tau, \varsigma_0, \dots, \varsigma_{c-2})$ , supplied by Theorem 5.18(7) and Section 5.8.2, transports the same statement to the original blowup coordinates.  $\square$

**Proposition 2.4** (QDM operation ledgers). *Let a marker fold be preserved by the regular scalar extensions and formal coordinate changes described above, and by every recorded grading suspension if its block type retains grading. The projective-bundle and blowup QDM comparisons on the even carriers then supply (2.1) and (2.2); each center marker is evaluated at its actual occurrence  $\omega_j$ .*

*Proof.* For a rank- $r$  bundle, Iritani–Koto’s Theorem 5.1 gives  $r$  copies of the base QDM after Laurent extension [9, Theorem 5.1]. Its invertible joint bulk Jacobian gives independent unit coordinates on the  $r$  target factors. In the  $j$ -th factor, changing its unit coordinate adds an independent scalar  $u_j I$  to Euler multiplication. For two finite spectra, the resultant of their translated characteristic polynomials is a nonzero polynomial in  $u_i - u_j$ , with leading coefficient one. Hence the spectra of distinct factors are disjoint at the generic point: their primary blocks do not fuse when the direct sum is formed. Regularity, even-carrier restriction, the fixed splitting field, and suspension invariance therefore carry the marker blockwise across the comparison, and additivity gives (2.1). Tensoring the bundle by a line bundle, when needed for the global-generation convention, changes neither the projective bundle nor the intrinsic Novikov embedding [9, Remark 1.2 and Remark 5.2].

For a codimension- $c$  blowup, Iritani’s Theorem 5.18 gives one ambient QDM and  $c - 1$  center QDMs [8, Theorem 5.18]. Parts (4), (6), and (7) give the comparison asymptotics, the leading reconstruction parameters, and the invertible combined bulk Jacobian, respectively. The independent unit coordinates supplied by part (7) give the same nonzero resultant argument, so the ambient and occurrence-indexed center spectra are pairwise disjoint generically. Regularity, even-carrier restriction, the fixed splitting field, suspension invariance, and Lemma 2.3 for each center copy then give (2.2). Each comparison is evaluated over its own common coefficient spine; the proof never composes Laurent comparisons recursively.  $\square$

The QDM comparison need not identify the two ledger elements before applying the fold. The relevant block-ledger operation is instead the additivity square

$$\begin{array}{ccc} \mathcal{L}_{\mathcal{T}} \times \mathcal{L}_{\mathcal{T}} & \xrightarrow{+} & \mathcal{L}_{\mathcal{T}} \\ w \times w \downarrow & & \downarrow w \\ A \times A & \xrightarrow{+} & A. \end{array} \quad (2.6)$$

At marker level, Proposition 2.4 supplies the resulting equalities in  $A$ ; its blowup statement also retains the occurrence metadata used by Theorem 2.2.

### 2.3 The rank-two formal-exponent marker

Let  $\mathcal{T}_{\text{at}}$  be the block type on the  $H^{\text{ev}}$ -carrier retaining the connection, pairing, and centered Euler endomorphism, with no absolute grading, marked bulk section, or loop framing, and specialize to  $A = \mathbf{N}$ . Let  $E$  be a generic even block with centered connection

$$z\partial_z y = A(z)y, \quad A(z) = z^{-1}N + A_0 + zA_1 + z^2A_2 + \cdots. \quad (2.7)$$

Assume  $\text{rk } E = 2$ ,  $N \neq 0$ , and  $N^2 = 0$ . Then  $L = \text{im } N = \ker N$  is a line. Define

$$E^\sharp = \{s \in E : s \bmod z \in L\}. \quad (2.8)$$

**Lemma 2.5** (The regular coefficient preserves the nilpotent line). *Horizontality of the Poincaré pairing implies  $A_0L \subset L$ .*

*Proof.* Separated spectral factors are orthogonal for the horizontal Poincaré pairing: coefficientwise horizontality kills every off-diagonal block by an invertible Sylvester operator. Hence the restriction to  $E$  is nondegenerate. Write it as  $P(z) = P_0 + zP_1 + \cdots$ . Horizontality gives

$$z\partial_z P(z) + A(z)^T P(z) + P(z)A(-z) = 0.$$

The  $z^{-1}$  coefficient gives  $N^T P_0 = P_0 N$ . Thus  $L$  is isotropic and, in dimension two,  $L^\perp = L$ . The constant coefficient is

$$A_0^T P_0 + P_0 A_0 + N^T P_1 - P_1 N = 0.$$

Evaluating this identity on  $(x, x)$ , where  $x \in L = \ker N$ , gives  $(A_0 x, x) = 0$ . Hence  $A_0 x \in L^\perp = L$ .  $\square$

Choose a frame with  $L = \langle e_1 \rangle$  and  $Ne_2 = \nu e_1$ ,  $\nu \neq 0$ . The modified lattice has frame  $e_1, ze_2$ , so

$$A^\sharp = S^{-1}AS - S^{-1}z\partial_z S, \quad S = \text{diag}(1, z). \quad (2.9)$$

The term  $z^{-1}N$  becomes regular. The only possible new pole is the  $E_{21}$ -coefficient of  $A_0$ , which vanishes by Lemma 2.5. Thus  $A^\sharp = R + zR_1 + \cdots$ . Define

$$\delta^\sharp(E) := (\text{tr } R)^2 - 4 \det R. \quad (2.10)$$

**Proposition 2.6** (Rank-two rigidity). *Under formal change of even bulk coordinates, the modified connection is regular in every base direction and*

$$\partial R = [G_\partial, R] \quad (2.11)$$

for a regular matrix  $G_\partial$ . Consequently  $\partial(\delta^\sharp) = 0$ . Regular connection isomorphisms preserve the domain and value of  $\delta^\sharp$ .

*Proof.* Write a centered base equation as  $\partial y = (z^{-1}C_\partial + C_{\partial,0} + O(z))y$ . The  $z^{-2}$  coefficient of flatness gives  $[N, C_\partial] = 0$ , and the trace of the  $z^{-1}$  coefficient gives  $\text{tr } C_\partial = 0$ ; here the base equation has been scalar-centered. The commutant of a nonzero rank-one nilpotent  $2 \times 2$  matrix is the coefficient-field span of  $I$  and  $N$ , so  $C_\partial = q_\partial N$ . The same  $z^{-1}$  coefficient gives

$$\partial N = -q_\partial N + [N, q_\partial A_0 - C_{\partial,0}].$$

Writing  $B = q_\partial A_0 - C_{\partial,0}$ , one has  $\partial(N^2) = -2q_\partial N^2 + [N^2, B]$ ; uniqueness for the formal coefficient recursion preserves  $N^2 = 0$ , while the nonzero leading value prevents  $N$  from



becoming the zero formal section. Thus formal bulk transport preserves the nonzero square-zero orbit and its image line. After modification, the only possible base pole is  $z^{-1}kE_{21}$ . If  $R = \begin{pmatrix} a & \nu \\ c & d \end{pmatrix}$ , the  $z^{-1}$  flatness equation is  $kE_{21} + [R, kE_{21}] = 0$ . Its diagonal entries are  $\nu k$  and  $-\nu k$ , so  $k = 0$ . The constant term of the flatness equation is now (2.11); trace and determinant are constant under infinitesimal conjugation. A regular isomorphism carries  $N$ ,  $L$ , and the modified lattice to their counterparts and conjugates the residue.  $\square$

**Proposition 2.7** (Exponent interpretation). *If  $\rho_1, \rho_2$  are the eigenvalues of  $R$ , their classes modulo  $\mathbf{Z}$  are the formal exponent classes of the original centered block and*

$$\delta^\sharp = (\rho_1 - \rho_2)^2.$$

*Proof.* The elementary modification uses integral powers of  $z$ , so it preserves formal exponent classes modulo  $\mathbf{Z}$ . The modified connection is regular singular with residue  $R$ ; the regular-singular classification identifies the exponent classes with the residue eigenvalues modulo  $\mathbf{Z}$  [11, Chapter 3]. The displayed identity is the quadratic discriminant formula.  $\square$

For such a block  $E$ , write  $\rho_1, \rho_2$  for the eigenvalues of its modified residue in the fixed algebraic closure  $\overline{K}$  of the generic coefficient field. Define the fold on each block by

$$w_{\text{at}}(E) = \begin{cases} 1, & \text{rk } E = 2, N \neq 0, N^2 = 0, [\rho_1] \neq [\rho_2] \text{ in } \overline{K}/\mathbf{Z}, \\ 0, & \text{otherwise,} \end{cases} \quad (2.12)$$

and extend it additively to  $w_{\text{at}} : \mathcal{L}_{\mathcal{T}_{\text{at}}} \rightarrow \mathbf{N}$ . Put  $I_{\text{at}}(Y) = w_{\text{at}}(\mathcal{B}_{\mathcal{T}_{\text{at}}}(Y))$ . Propositions 2.6 and 2.7 make this a lawful marker for Proposition 2.4: conjugation preserves the two residue eigenvalues, while changes of regular-singular lattice and recorded suspensions shift exponents only by integers. The condition  $\delta^\sharp \neq 0$  is insufficient: it says only  $\rho_1 \neq \rho_2$  and therefore also marks resonant blocks.

## 2.4 The cubic block

Let  $X \subset \mathbf{P}^4$  be a smooth cubic threefold, let  $P = H|_X$ , and put  $q = Q^\ell$  for the line class and  $r = (3q)^{1/2}$ . Beauville's formulas give [3, main theorem and (2.1)–(2.3)]

$$P \star P = P^2 + 6q, \quad P \star P^2 = P^3 + 15qP, \quad P \star P^3 = 6qP^2 + 36q^2. \quad (2.13)$$

Thus, in the basis  $(1, P, P^2, P^3)$ ,

$$K_X = 2 \begin{pmatrix} 0 & 6q & 0 & 36q^2 \\ 1 & 0 & 15q & 0 \\ 0 & 1 & 0 & 6q \\ 0 & 0 & 1 & 0 \end{pmatrix}, \quad G_X = \frac{1}{2} \text{diag}(3, 1, -1, -3), \quad (2.14)$$

and  $z^2 \partial_z S = (K_X + zG_X)S$ . This agrees with [5, Section 3].

**Proposition 2.8** (Small-even cubic block reduction). *The characteristic and minimal polynomials of  $K_X$  are both  $T^2(T^2 - 108q)$ . After adjoining  $r$ , its eigenvalues are  $6r, -6r, 0, 0$ , and the zero generalized eigenspace is one rank-two Jordan block. A normalized gauge regular in  $z$  gives*

$$\begin{aligned} z^2 \partial_z \tilde{S}_+ &= (6r + O(z^2)) \tilde{S}_+, \\ z^2 \partial_z \tilde{S}_- &= (-6r + O(z^2)) \tilde{S}_-, \\ z^2 \partial_z \begin{pmatrix} \tilde{S}_3 \\ \tilde{S}_4 \end{pmatrix} &= [J_0 + zD_0 + z^2E_0 + O(z^3)] \begin{pmatrix} \tilde{S}_3 \\ \tilde{S}_4 \end{pmatrix}, \end{aligned} \quad (2.15)$$

where

$$J_0 = \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix}, \quad D_0 = \begin{pmatrix} -19/18 & 0 \\ 0 & 19/18 \end{pmatrix}, \quad E_0 = \begin{pmatrix} 0 & -14/(81r^2) \\ -8/81 & 0 \end{pmatrix}. \quad (2.16)$$

*Proof.* The matrix

$$C = \begin{pmatrix} 6r^3 & -6r^3 & 0 & -7r^2 \\ 7r^2 & 7r^2 & -2r^2 & 0 \\ 3r & -3r & 0 & 1 \\ 1 & 1 & 1 & 0 \end{pmatrix}, \quad \det C = -486r^5,$$

puts  $K_X$  into

$$C^{-1}K_X C = \begin{pmatrix} 6r & 0 & 0 & 0 \\ 0 & -6r & 0 & 0 \\ 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Partition as  $1 \mid 1 \mid 2$ . At each positive order in  $z$ , the off-diagonal gauge coefficient is the unique solution of a Sylvester equation between blocks with disjoint scalar spectra. The gauge and inverse are regular in  $z$ ; explicitly, it has the normalized form

$$A(z) = I + \sum_{n \geq 1} A_n z^n, \quad A_n \text{ block-off-diagonal.} \quad (2.17)$$

Direct multiplication through order  $z^2$  gives (2.16).  $\square$

For the zero block,  $N = J_0 = 2E_{12}$ . With  $S = \text{diag}(1, z)$ ,

$$R_X = 2E_{12} + D_0 - \frac{8}{81}E_{21} - \text{diag}(0, 1) = \begin{pmatrix} -19/18 & 2 \\ -8/81 & 1/18 \end{pmatrix}. \quad (2.18)$$

Its characteristic polynomial is  $(T + 1/6)(T + 5/6)$ . Thus the two exponent classes differ by  $2/3 \notin \mathbf{Z}$ , so the block is marked; as a numerical check,

$$\boxed{\delta^\#(X) = \frac{4}{9}}, \quad I_{\text{at}}(X) = 1. \quad (2.19)$$

## 2.5 Vanishing on centers through dimension two

**Proposition 2.9** (Vanishing on low-dimensional centers). *For every smooth projective  $Z$  of dimension at most two and every actual QDM comparison occurrence  $\omega$ ,  $I_{\text{at}}(Z; \omega) = 0$ .*

*Proof.* A point has one rank-one block. For  $\mathbf{P}^1$ , the relation  $p \star p = Qe^{t_p}$  gives two rank-one generic blocks.

Let  $C$  have genus  $g \geq 1$ . There is no nonconstant genus-zero map to  $C$ , so its even big quantum product is classical. With  $\int_C p = 1$  and  $a = 2 - 2g$ , its centered even connection is

$$z\partial_z Y = \left[ z^{-1}aE_{21} + \text{diag}(1/2, -1/2) \right] Y.$$

For  $g = 1$ ,  $N = 0$ . For  $g > 1$ , the modification  $S = \text{diag}(z, 1)$  gives residue  $aE_{21} - \frac{1}{2}I$ , of discriminant zero. Its two eigenvalues, and therefore its two exponent classes, are equal. Thus every curve has zero marker.

Let  $S$  be a surface with  $K_S$  nef. For a quantum-product term with input degree  $b$ , non-scalar Euler degree  $a \geq 2$ , curve class  $d$ , and nonunit even bulk degrees  $e_\nu \geq 2$ , the dimension axiom gives output degree

$$a + b - 2c_1(S) \cdot d + \sum_{\nu} (e_\nu - 2) \geq b + 2.$$

The centered Euler operator strictly raises ordinary degree. Its only block is the whole even cohomology, of rank  $b_0 + b_2 + b_4 \geq 3$ , so its marker is zero.

A minimal surface with non-nef canonical class is  $\mathbf{P}^2$  or a ruled surface  $\mathbf{P}_C(V)$  [4, Chapter VI]. The relation  $H^{*3} = Q$  makes  $\mathbf{P}^2$  generically semisimple. Formula (2.1) and the curve case give zero for ruled surfaces. Every nonminimal surface is obtained from a minimal one by point blowups; formula (2.2) adds only zero point terms. This proves intrinsic nullity. Lemma 2.3 identifies every faithfully tagged actual center occurrence with a faithful scalar extension of the corresponding intrinsic generic block, and the marker is unchanged by that extension.  $\square$

## 2.6 The projective endpoint and the contradiction

Formula (2.1) applied to  $X \times \mathbf{P}^1 = \mathbf{P}_X(\mathcal{O}_X^{\oplus 2})$  gives

$$I_{\text{at}}(X \times \mathbf{P}^1) = 2. \quad (2.20)$$

On the other hand, over the generic numerical Novikov field  $\Lambda = \text{Frac } \mathbf{C}[[Q]]$ ,  $QH^\bullet(\mathbf{P}^4) = \Lambda[H]/(H^5 - Q)$ . At  $Q \neq 0$ , multiplication by  $5H$  has five distinct eigenvalues after algebraic closure. Thus

$$I_{\text{at}}(\mathbf{P}^4) = 0. \quad (2.21)$$

*Proof of Theorem 1.1.* Proposition 2.9 kills every actual center correction in a weak factorization of smooth projective fourfolds. Hence Theorem 2.2 makes  $I_{\text{at}}$  birationally invariant. If  $X \times \mathbf{P}^1$  were rational, then

$$2 = I_{\text{at}}(X \times \mathbf{P}^1) = I_{\text{at}}(\mathbf{P}^4) = 0,$$

a contradiction.  $\square$

## 3 Consequences

*Proof of Corollary 1.2.* In dimension three, every center in projective weak factorization has dimension at most one. The curve part of Proposition 2.9 therefore makes  $I_{\text{at}}$  birationally invariant. The generic quantum product of  $\mathbf{P}^3$  is semisimple, so  $I_{\text{at}}(\mathbf{P}^3) = 0$ . A marked block gives  $I_{\text{at}}(Y) > 0$ , ruling out rationality.  $\square$

*Proof of Corollary 1.3.* Kuznetsov's flop gives a birational map between rank-two projective bundles  $\mathbf{P}_V(U)$  and  $\mathbf{P}_X(E^*)$  over  $V$  and its associated Pfaffian cubic threefold  $X$  [10, Theorems 2.17–2.18]. Each projective bundle is birational to the product of its base with  $\mathbf{P}^1$ . Hence  $V \times \mathbf{P}^1$  is birational to  $X \times \mathbf{P}^1$ , which is irrational by Theorem 1.1.  $\square$

*Proof of Corollary 1.4.* Voisin produces a nonempty countable union of subvarieties of codimension at most three in the moduli space of smooth cubic threefolds along which the primitive minimal class  $\Theta^4/4!$  is algebraic, hence along which  $CH_0$  is universally trivial [12, Theorem 4.5 and Lemma 4.6]. The projective-bundle formula passes universal  $CH_0$ -triviality to the product, and Theorem 1.1 supplies its irrationality.  $\square$

*Proof of Corollary 1.5.* The Fermat equation is a sum of five forms in separated single variables, so [7] gives universal  $CH_0$ -triviality; the projective-bundle formula passes it to the product. Irrationality follows from Theorem 1.1.  $\square$

*Proof of Corollary 1.6.* The cubics of [13, Theorem 3.3] carry unirational parametrizations of degrees two and three, and [13, Corollary 3.5] keeps both degrees on  $X \times \mathbf{P}^1$ . Parametrizations of coprime degrees give a decomposition of the diagonal and hence universal  $CH_0$ -triviality [13, Section 1]. Theorem 1.1 gives irrationality.  $\square$

The marker operation formula has the same additive shape as Bittner’s blowup relation [2, Theorem 3.1]. More precisely, if an additive extension of a marker is defined on the Bittner group, then

$$I(\mathrm{Bl}_Z Y) - I(\mathbf{P}_Z(N_{Z/Y})) = I(Y) - I(Z)$$

follows from  $I(\mathrm{Bl}_Z Y) = I(Y) + (c - 1)I(Z)$  and  $I(\mathbf{P}_Z(N_{Z/Y})) = cI(Z)$ . The effective  $\mathbf{N}$ -valued fold used here is not itself such a group-valued motivic measure: no subtraction or multiplicativity has been constructed. The parallel explains why the center bound is decisive without claiming a Grothendieck-ring factorization that the proof does not supply.

All results above concern one product stabilization. No statement about a second or arbitrary stabilization is implicit.

## AI assistance disclosure

This project was developed with extensive assistance from OpenAI Codex and Anthropic Claude in proof exploration, literature research, exact computation, formal verification, and drafting. AI-assisted adversarial review identified, in particular, the need for the even-carrier restriction and for replacing the preliminary nonzero-discriminant test by distinct formal exponent classes in Section 2. The author checked the arguments, computations, code, and cited sources, reviewed and edited all AI-assisted material, and assumes responsibility for all content.

Numbered statements are cited from the exact versions listed here. Where both a journal and a preprint version appear, the numbering is that of the preprint.

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